Jia Hao Liang (Frank)

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EDUCATION

2026-2029 **MASSACHUSETTS INSTITUTE OF TECHNOLOGY** Cambridge, MA
Major: Artificial Intelligence and Decision Making (Course 6-4) & Mathematics (Course 18)

2025-2026 **COLUMBIA UNIVERSITY, COLUMBIA COLLEGE** New York, NY
(Transferred) **GPA:** 4.16/4.00; **Courses:** Honors Math A & B (A+), Probability Theory (A+), Intro to Quantum Computing (A+), Modern Algebra I (A+), Data Structures (A), Literature Humanities I & II (A)

- **Fellowship:** Science Research Fellow (Columbia Named Scholars, CUSP); Dean's list all term

RESEARCH & EXPERIENCE

2025-2026 **SOFTWARE SYSTEMS LABORATORY (SSL)** Columbia University
Researcher (Advised by Prof. Junfeng Yang & Ph.D. Student Yun-Yun Tsai)

- Investigating fingerprinting methods for distributed agentic frameworks to defend multi-agent LLM systems against adversarial theft; Science Research Fellow Summer Program
- Developed RAFP, a rare-region adversarial fingerprinting method for LLM lineage verification, establishing theoretical robustness guarantees (preprint on arXiv)

2026 **MACHINE LEARNING PASSION PROJECT** Massachusetts Institute of Technology
Researcher

- Built Transformer-based language model from scratch, implementing a BPE tokenizer, Attention blocks, and Decoder. Published trained model weights on HuggingFace
- Profiled forward/backward/optimizer performance, implemented a fused FlashAttention-2 Triton kernel, and distributed training across GPUs with DDP, optimizer sharding, and FSDP

2026 **DIRECTED READING PROGRAM** Columbia University
Topic: Algorithmic Game Theory (Advised by Ph.D. student Aditya Shankar Garg)

- Studied computational complexity of game theory algorithms, e.g. Nash Equilibrium

2024 **YAU MATHCAMP** Fudan University, Shanghai
Best Project Award, Group Leader

- Studied Galois Theory and conducted research on the Inverse Galois Problem
- Selected to present research findings at the concluding Mathcamp colloquium (5 out of 70)
- Listened to seminars from Fields Medalist Shing-Tung Yau and Andrei Okounkov

2023-2025 **MACAU INTELLIGENT BUS PROGRAM** Macau University of Science and Technology
Developer, Third Prize

- Developed a mobile application that integrated variables such as real-time bus location, applying machine learning and algorithmic optimization to compute efficient commuter routes
- Awarded Third Prize in the Youth AI and Future Media Global Innovation Contest

HONORS & AWARDS

2025 **INTERNATIONAL MATHEMATICAL OLYMPIAD ; CHINESE MATHEMATICAL OLYMPIAD**
Honorable Mention ; Bronze Medal

- Best Performer in Macau for Macau Team Selection Test, IMO, China MO, and China TST

2023 & 2024 **ASIAN PHYSICS OLYMPIAD ; CHINESE PHYSICS OLYMPIAD**
Bronze Medal ; Bronze Medal x2

- Best Performer in Macau Team (qualified for IPhO 2024, Macau did not attend due to location)

2024 **INTERNATIONAL OLYMPIAD IN ARTIFICIAL INTELLIGENCE**
Bronze Medal

- Applied large language models and ML algorithms to problems in Google Colab, gaining hands-on experience across Machine Learning, Computer Vision, and NLP

2026 **COLUMBIA MATH PRIZE EXAM**
Professor Van Amringe Mathematical Prize (single freshman recipient)

- Most proficient in designated mathematical subjects based on a special departmental exam

2026 **COLUMBIA INTEGRATION BEE**
Grand Integrator (First freshman champion in the competition's history)

LEADERSHIP

2025-2026 **COLUMBIA JOURNAL OF UNDERGRADUATE MATHEMATICS; COLUMBIA SCIENCE REVIEW; COLUMBIA UNDERGRADUATE SCIENCE JOURNAL** Columbia University, NYC
Board Member; Writer; Editor

- Published article about Knot Theory and Game Theory in Columbia Science Review

2021-2025 **PUI CHING MIDDLE SCHOOL MATH CLUB** Pui Ching Middle School, Macau
President

- Instructed ~30 club members, hosted Math competitions, delivered Math talks

SKILLS

- Machine Learning, Python, PyTorch, C++, HuggingFace, Quantum Computing
- Math Modeling, Probability & Statistics, MATLAB, LaTeX, Game Theory, Physics
- Influential Transportation Content Creator ("flbjcy" on Bilibili, 21.5k+ views)